

Course Directive



ID607001: Introductory Application Development Concepts

Course Information

Level	Credits	Prerequisite	Timetable
6	15	ID511001: Programming 2	Monday 1:00 PM - 2:45 PM in D202 and Thursday 1:00 PM - 2:45 PM in D202

Teaching Staff

Name	Position	Office Location	Email Address
Grayson Orr	Principal Lecturer	D309	grayson.orr@op.ac.nz

Course Dates

Term 1 (7 weeks)	Mid-Semester Break	Term 2 (9 weeks)
23 February - 10 April	11 April - 27 April	28 April - 03 July

Aims

To introduce the concepts of application development including algorithms, data structures and design patterns that are required to use a simple, industry-relevant development framework.

Learning Outcome

At the successful completion of this course, learners will be able to:

1. Design and build secure applications with dynamic database functionality following an appropriate software development methodology.

Assessments

Assessment	Weighting	Due Date	Learning Outcome
Practical	20%	22 May at 4.59 PM	1
Project	80%	26 June at 4.59 PM	1

Grade Table - Criterion Referenced

Grade	Mark Range
A+	Met all course requirements - mark in range [90-100]
A	Met all course requirements - mark in range [85-89]
A-	Met all course requirements - mark in range [80-84]
B+	Met all course requirements - mark in range [75-79]
B	Met all course requirements - mark in range [70-74]
B-	Met all course requirements - mark in range [65-69]
C+	Met all course requirements - mark in range [60-64]
C	Met all course requirements - mark in range [55-59]
C-	Met all course requirements - mark in range [50-54]
D	There at end. Did not meet course requirements. Mark in range [40-49]
E	There at end. Did not meet course requirements. Mark in range [0-39]

Provisional Schedule

Week	Topics
1/Tahi	Git and JavaScript
2/Rua	API, Express and Development Tools
3/Toru	PostgreSQL, Docker, ORM, JSDoc and Postman
4/Whā	Content Negotiation, Relationships and N-Layer Architecture
5/Rima	Validation, Seeding, Query Parameters and Deployment
6/Ono	Security, Authentication and Role-Based Access Control
7/Whitu	Backend Testing (Integration), Code Coverage, CI/CD and GitHub Actions
	Mid Semester Break

Week	Topics
8/Waru	Vite and SvelteKit.js Design
9/Iwa	API Integration, Bootstrap and Deployment
10/Tekau	Performance and UI/UX Design
11/Tekau mā tahi	Frontend Testing (Component and End-to-End)
12/Tekau mā rua	Assessment Work
13/Tekau mā toru	Assessment Work
14/Tekau mā whā	Assessment Work
15/Tekau mā rima	Assessment Work
16/Tekau mā ono	Assessment Marking

Resources

Software

This course will be taught using [Microsoft Visual Studio Code](#) and [Node.js](#).

Readings

No textbook is required for this course. URLs to useful resources will be provided in the lecture notes.

Course Requirements and Expectations

Learning Hours

This course requires 150 hours of learning. This time includes 60 hours of directed learning hours and 90 of self-directed learning hours.

Criteria for Passing

To pass this course, you must achieve a cumulative pass mark of 50% over all assessments. There are no reassessments or resits.

Attendance

- Learners are expected to attend all classes, including lectures and labs.
- If you cannot attend for a few days for any reason, contact the course.

Communication

Microsoft Outlook/Teams are the official communication channels for this course. It is your responsibility to regularly check Microsoft Outlook/Teams and [GitHub](#) for important course material, including changes to class scheduling or assessment details. Not checking will not be accepted as an excuse.

Snow Days/Polytechnic Closure

In the event Otago Polytechnic is closed or has a delayed opening because of snow or bad weather, you should not attempt to attend class if it is unsafe to do so. It is possible that the teaching staff will not be able to attend either, so classes will not physically meet. However, this does not become a holiday. Rather, the course material will be made available on [GitHub](#) for classes affected by the closure. You are responsible for any course material presented in this manner. Information about closure will be posted on the [Otago Polytechnic Facebook](#) page.

Group Work and Originality

Learners in the Bachelor of Information Technology programme are expected to hand in original work. Learners are encouraged to discuss assessments with their fellow learners. However, all assessments are to be completed as individual works unless group work is explicitly required (i.e. if it doesn't say it is group work, then it is not group work - even if a group consultation was involved). Failure to submit your original work will be treated as plagiarism.

AI Tools

Learning to use AI tools is an important skill. While AI tools are powerful, you must be aware of the following:

- If you provide an AI tool with a prompt that is not refined enough, it may generate a not-so-useful response
- Do not trust the AI tool's responses blindly. You must still use your judgement and may need to do additional research to determine if the response is correct
- Acknowledge what AI tool you have used. In the assessment's repository README.md file, please include what prompt(s) you provided to the AI tool and how you used the response(s) to help you with your work

Referencing

Appropriate referencing is required for all work. Referencing standards will be specified by the teaching staff.

Plagiarism

Plagiarism is submitting someone else's work as your own. Plagiarism offences are taken seriously, and an assessment that has been plagiarised may be awarded a zero mark. A definition of plagiarism is in the Student Handbook, available online or at the school office.

Submission Requirements

All assessments are to be submitted by the time, date and method given when the assessment is issued. Failure to meet all requirements will result in a penalty of up to 10% per day, including weekends.

Extensions

Familiarise yourself with the assessment due dates. Extensions will only be granted if you are unable to complete the assessment by the due date because of unforeseen circumstances outside your control. The length of the extension granted will depend on the circumstances and must be negotiated with the teaching staff before the assessment due date. A medical certificate or support letter may be needed. Extensions will not be granted for poor time management or the pressure of other assessments.

Impairment

In case of sickness, contact the teaching staff or Head of School as soon as possible, preferably before the assessment is due. The policy regarding the granting of a mark that considers impaired performance requires a medical certificate and a medical practitioner's signature on a form. You may refer to the guide on impaired performance in the student handbook.

Appeals

If you are concerned about any aspect of your assessment, approach the teaching staff in the first instance. We support an open-door policy and aim to resolve issues promptly. Further support is available from the Head of School. Otago Polytechnic has a formal process for academic appeals if necessary.

Other Documents

Regulatory documents relating to this course can be found on the [Otago Polytechnic](#) website.

Author: Grayson Orr

Course: ID607001: Introductory Application Development Concepts